

Which UPS OR INVERTER?

Your guide to the best inverter-battery combination that gives you the best backup but least headache



Luminous long-backup UPS

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Whenver power supply fails and you sweat, you need an inverter. So an inverter is a need-based product, not a 'want' product.

Basically, an inverter system comprises an inverter, charger and battery. When mains power is available, it converts AC into DC and stores the same in the battery. When mains supply fails, it inverts the DC power stored

in the battery into AC to run the connected appliances.

Before you buy an inverter for your home, there are several things to consider, like:

1. Do I need a built-in UPS?
2. Which type of inverter (squarewave, modified sinewave or pure sinewave) will suit me?
3. How much inverter capacity do I need for running the desired loads?
4. How long can I run the appliances on the inverter? (This depends on the battery size (in ampere-hours) and the type of batteries used.)
5. Protection features
6. Additional features offered

by the inverter, such as battery level (in per cent) display, load display, auto-adjustment to battery and no-load shutdown

Here are some recommendations to help you choose the best inverter for you.

Do I need a UPS model?

Inverters with built-in UPS come with over- and under-voltage protection. So in the event of wide voltage fluctuations, your PC isn't affected. Also, changeover time from mains to battery is fast and therefore your PC

doesn't boot.

It's better to go for the UPS model because even if you don't have a computer at home, you may buy one tomorrow. Also, inverters with UPS are available for more or less the same price as standalone inverters.

Luminous, for instance, has inverters up to 6 kVA with built-in UPS. All Parker GPH Home UPS up to 3.5 kVA have built-in UPS.

Which type of inverter will suffice?

Inverters convert AC into DC and then DC into AC, which can be created as square wave, modified sine wave or pure sine wave based on the filtering process.

All squarewave inverters today are more or less modified sinewave type. Modified sinewave inverters are somewhere in between pure squarewave and sinewave. The difference is only in terms of the waveshape. People have been using squarewave and modified sinewave without any problem *per se*. Square wave doesn't affect the life of appliances unless it's a sensitive equipment.

In sinewave, you get a waveshape similar to that of AC mains. It doesn't produce the humming noise from a fan or tubelight that's produced with a squarewave inverter. For sensitive loads like audio appliances (CD players and home theatre systems), water motors and ACs, pure sinewave is recommended.

Sinewave is not a new technol-